

CFA Level I

Quantitative Methods

Practice Questions

Practice Question Book

220 CFA-style multiple-choice questions
20 questions across each of the 11 learning modules

Grounded in the 2025 CFA Program Curriculum, Level I, Volume 1 - Quantitative Methods. For personal exam preparation and practice.

Module 1: Rates and Returns

20 questions

- Q1.** An interest rate is BEST described as having which three possible interpretations?
- A. Required rate of return, discount rate, and opportunity cost
 - B. Inflation premium, default premium, and liquidity premium
 - C. Nominal rate, real rate, and risk-free rate
- Q2.** An interest rate can be viewed as the sum of the real risk-free rate and a set of premiums. Which of the following is NOT one of those premiums?
- A. Inflation premium
 - B. Default risk premium
 - C. Equity risk premium
- Q3.** The nominal risk-free interest rate is BEST approximated as the sum of the:
- A. real risk-free rate and the inflation premium
 - B. real risk-free rate and the default risk premium
 - C. real risk-free rate and the maturity premium
- Q4.** An investor buys a stock for 50, receives a dividend of 2, and sells it for 54. The holding period return is closest to:
- A. 8.0%
 - B. 12.0%
 - C. 4.0%
- Q5.** Which return measure is LEAST sensitive to the timing and amount of cash flows into and out of a portfolio?
- A. Money-weighted return
 - B. Time-weighted return
 - C. Internal rate of return
- Q6.** The time-weighted rate of return is generally the PREFERRED measure for evaluating portfolio managers because:
- A. it accounts for the size of the investor's cash flows
 - B. external cash flows are generally outside the manager's control
 - C. it always exceeds the money-weighted return
- Q7.** A money-weighted rate of return is BEST described as the:
- A. compound growth rate of one unit of currency invested
 - B. internal rate of return on a portfolio accounting for all cash flows
 - C. simple average of subperiod returns
- Q8.** For a return series of +50% followed by -50%, the geometric mean return is closest to:
- A. 0%
 - B. -13.4%
 - C. +13.4%
- Q9.** Relative to the arithmetic mean, the geometric mean return of a volatile, non-constant return series is:
- A. always greater
 - B. always less

C. always equal

- Q10.** Which mean is MOST appropriate for averaging ratios such as price-to-earnings multiples to reduce the impact of large outliers?
- A. Harmonic mean
 - B. Arithmetic mean
 - C. Geometric mean
- Q11.** A trimmed mean is BEST described as a mean computed after:
- A. replacing the extreme values with the nearest retained values
 - B. excluding a stated small percentage of the lowest and highest values
 - C. weighting observations by their probability
- Q12.** A winsorized mean differs from a trimmed mean in that the winsorized mean:
- A. excludes the extreme observations entirely
 - B. replaces the extreme observations with specified boundary values
 - C. uses only the median observation
- Q13.** A quarterly return of 3% compounded over four quarters produces an annualized return closest to:
- A. 12.00%
 - B. 12.55%
 - C. 11.45%
- Q14.** As the number of compounding periods per year approaches infinity, interest is said to be:
- A. compounded continuously
 - B. compounded discretely
 - C. simple interest
- Q15.** If the holding period return over a stated period is 6%, the continuously compounded return for that period is closest to:
- A. 5.83%
 - B. 6.18%
 - C. 6.00%
- Q16.** Gross return is BEST described as the return:
- A. after deducting all managerial and administrative expenses
 - B. prior to deduction of managerial and administrative expenses not directly related to return generation
 - C. after deducting taxes on dividends and gains
- Q17.** Net return is BEST described as:
- A. gross return less managerial and administrative expenses
 - B. gross return less only trading costs
 - C. gross return plus the inflation premium
- Q18.** The after-tax nominal return is computed as the total return minus:
- A. the inflation premium
 - B. any allowance for taxes on dividends, interest, and realized gains
 - C. managerial expenses only
- Q19.** Real returns are PARTICULARLY useful for:
- A. comparing returns across time periods when inflation rates vary
 - B. evaluating a portfolio manager's stock selection skill
 - C. measuring returns before any expenses

- Q20.** Leveraging a portfolio through borrowing or futures will MOST likely:
- A. reduce both expected gains and losses
 - B. amplify both the portfolio's gains and losses
 - C. have no effect on the portfolio's risk

Module 2: Time Value of Money in Finance

20 questions

- Q21.** The price of a coupon bond is BEST described as the sum of the present values of the bond's:
- A. promised coupon payments and its par value
 - B. par value only
 - C. coupon payments only
- Q22.** For a pure discount (zero-coupon) bond, the price reflects the present value of:
- A. the coupon payments only
 - B. the par value only
 - C. the coupons plus par value
- Q23.** According to the dividend discount model, the value of a stock should reflect the sum of the present values of:
- A. the stock's expected future dividends in perpetuity
 - B. the stock's expected future earnings
 - C. the stock's book value plus next year's dividend
- Q24.** Stock valuation models based on dividends are typically classified by the assumed:
- A. expected growth pattern of future dividends
 - B. frequency of dividend payments
 - C. industry of the issuer
- Q25.** Using the constant-growth (Gordon) model, a stock with an expected dividend of 2.00, required return of 10%, and growth of 4% has a value closest to:
- A. 20.00
 - B. 33.33
 - C. 50.00
- Q26.** A no-growth (zero-growth) stock paying a constant dividend is valued as a:
- A. perpetuity: D / r
 - B. growing perpetuity: $D_1 / (r - g)$
 - C. finite annuity
- Q27.** Using the Gordon growth model, a stock's required return can be expressed as:
- A. $(D_1 / P_0) + g$
 - B. $(D_1 / P_0) - g$
 - C. $g - (D_1 / P_0)$
- Q28.** A stock trades at 40, has an expected dividend of 1.60, and a required return of 9%. Its implied dividend growth rate is closest to:
- A. 5.0%
 - B. 4.0%
 - C. 9.0%
- Q29.** If a bond's price is known, the bond's implied return is BEST found by solving for the:
- A. rate that equates the present value of cash flows to the price
 - B. coupon rate divided by par value
 - C. current yield only

- Q30.** The cash flow additivity principle allows two or more cash flow streams to be compared as long as the cash flows:
- A. occur at the same point in time
 - B. have the same nominal value
 - C. are from the same issuer
- Q31.** The cash flow additivity principle is MOST directly important because it ensures:
- A. the no-arbitrage condition holds
 - B. all bonds have equal yields
 - C. inflation is properly accounted for
- Q32.** Application of cash flow additivity confirms that asset prices are the same for:
- A. economically equivalent assets, even with differing cash flow streams
 - B. only assets with identical cash flow streams
 - C. only assets from the same market
- Q33.** An implied forward interest rate is MOST directly derived using:
- A. the cash flow additivity / no-arbitrage principle
 - B. the dividend discount model
 - C. the coefficient of variation
- Q34.** The future value of a single cash flow under continuous compounding is given by:
- A. $FV = PV(1 + r)^t$
 - B. $FV = PV \times e^{(rt)}$
 - C. $FV = PV / (1 + r)^t$
- Q35.** A 1,000 par, 5% annual coupon, 3-year bond priced when the market discount rate is 5% will trade at:
- A. a premium to par
 - B. par value (1,000)
 - C. a discount to par
- Q36.** All else equal, if the required return on a bond rises above its coupon rate, the bond will trade:
- A. at a premium
 - B. at a discount
 - C. at par
- Q37.** In the constant-growth dividend model, holding D_1 and r constant, a higher growth rate g results in:
- A. a higher stock value
 - B. a lower stock value
 - C. no change in value
- Q38.** Cash flow additivity can be applied to value all of the following EXCEPT:
- A. implied forward exchange rates
 - B. option values via replicating portfolios
 - C. the skewness of a return distribution
- Q39.** A stock is valued using a two-stage (changing growth) dividend model when:
- A. dividends are expected to grow at a high rate then settle to a stable rate
 - B. the firm never pays dividends
 - C. the discount rate is zero
- Q40.** The present value of a cash flow under continuous discounting is given by:
- A. $PV = FV \times e^{(-rt)}$

B. $PV = FV \times e^{(rt)}$

C. $PV = FV \times (1 + r)^t$

Module 3: Statistical Measures of Asset Returns

20 questions

- Q41.** Which measure of central tendency is MOST useful for skewed distributions because it is not influenced by extreme values?
- A. Arithmetic mean
 - B. Median
 - C. Mode
- Q42.** The only measure of central tendency that can be used with nominal or categorical data is the:
- A. mean
 - B. median
 - C. mode
- Q43.** A distribution with two modes is described as:
- A. unimodal
 - B. bimodal
 - C. trimodal
- Q44.** Quartiles, quintiles, deciles, and percentiles are collectively known as:
- A. measures of dispersion
 - B. quantiles
 - C. measures of skewness
- Q45.** In a box-and-whiskers plot, the 'box' depicts the:
- A. interquartile range ($Q3 - Q1$)
 - B. full range (max - min)
 - C. standard deviation
- Q46.** The range is considered to have limited usefulness as a dispersion measure because it:
- A. uses information from only two observations
 - B. cannot be computed for financial data
 - C. is always negative
- Q47.** When computing a SAMPLE variance, the sum of squared deviations is divided by:
- A. n
 - B. $n - 1$
 - C. $n + 1$
- Q48.** The mean absolute deviation (MAD) for a sample is the average of the:
- A. squared deviations of observations from the mean
 - B. absolute deviations of observations from the mean
 - C. deviations of observations from the median
- Q49.** A risk measure that focuses only on returns falling below a specified target is the:
- A. coefficient of variation
 - B. target downside deviation
 - C. range
- Q50.** The coefficient of variation (CV) is BEST described as:
- A. the standard deviation divided by the mean

- B. the mean divided by the standard deviation
- C. the variance minus the mean

Q51. A key advantage of the coefficient of variation is that it:

- A. has units of currency
- B. is a scale-free measure allowing direct comparison across datasets
- C. ignores the mean entirely

Q52. Two portfolios have equal standard deviations, but Portfolio A has a higher mean return. Comparing coefficients of variation, Portfolio A has:

- A. a lower CV (less risk per unit of return)
- B. a higher CV
- C. the same CV

Q53. An asset return distribution with frequent small losses and a few extreme gains relative to a normal distribution exhibits:

- A. positive skewness
- B. negative skewness
- C. zero skewness

Q54. A return distribution with frequent small gains and a few extreme losses is BEST described as:

- A. positively skewed
- B. negatively skewed
- C. symmetric

Q55. The kurtosis of a normal distribution is:

- A. 0
- B. 1
- C. 3

Q56. A distribution with fatter tails than the normal distribution is described as:

- A. leptokurtic (fat-tailed)
- B. platykurtic (thin-tailed)
- C. mesokurtic

Q57. Excess kurtosis greater than zero indicates a distribution that is:

- A. thin-tailed relative to normal
- B. fat-tailed relative to normal
- C. identical to the normal distribution

Q58. For investment returns, a leptokurtic (fat-tailed) distribution implies that extreme outcomes are:

- A. less likely than under a normal distribution
- B. more likely than under a normal distribution
- C. equally likely as under a normal distribution

Q59. The correlation coefficient between two variables ranges between:

- A. 0 and +1
- B. -1 and +1
- C. -infinity and +infinity

Q60. A correlation coefficient close to zero between two asset returns indicates:

- A. a strong positive linear relationship
- B. little or no linear relationship

C. a strong negative linear relationship

Module 4: Probability Trees and Conditional Expectations

20 questions

- Q61.** The expected value of a random variable X is BEST described as the:
- A. probability-weighted average of the possible outcomes
 - B. simple average of the possible outcomes
 - C. most likely single outcome
- Q62.** The variance of a random variable X is defined as:
- A. $E[X - E(X)]$
 - B. $E\{[X - E(X)]^2\}$
 - C. the square root of $E(X)$
- Q63.** A stock has a 60% probability of a 10% return and a 40% probability of a -5% return. Its expected return is closest to:
- A. 4.0%
 - B. 2.5%
 - C. 7.0%
- Q64.** Standard deviation is preferred to variance for interpreting dispersion because it is:
- A. always smaller than the variance
 - B. measured in the same units as the variable
 - C. unaffected by outliers
- Q65.** A probability tree is BEST described as a means of illustrating:
- A. the results of two or more events and their probabilities
 - B. a single deterministic outcome
 - C. the correlation between two variables
- Q66.** The conditional probability of event A given event B , $P(A|B)$, is defined as:
- A. $P(AB) / P(B)$, with $P(B)$ not equal to 0
 - B. $P(A) \times P(B)$
 - C. $P(A) + P(B) - P(AB)$
- Q67.** Two events A and B are independent if and only if:
- A. $P(A|B) = P(A)$
 - B. $P(A|B) = P(B)$
 - C. $P(AB) = P(A) + P(B)$
- Q68.** According to the total probability rule, if $S_1, S_2, \dots, S_n$ are mutually exclusive and exhaustive, then $P(A)$ equals:
- A. the sum of $P(A | S_i) \times P(S_i)$ across all scenarios
 - B. the product of all $P(S_i)$
 - C. $P(A | S_1)$ only
- Q69.** Bayes' formula is BEST described as a method to:
- A. update prior probabilities based on new information
 - B. compute the variance of a portfolio
 - C. rank data without distributional assumptions
- Q70.** In Bayes' formula, the updated probability equals the prior probability multiplied by the ratio of:

- A. probability of the new information given the event to the unconditional probability of the new information
- B. the unconditional probability of the event to the conditional probability
- C. the variance to the mean

Q71. Conditional expected value $E(X|S)$ is calculated as the probability-weighted average of outcomes:

- A. using conditional probabilities given scenario S
- B. ignoring scenario S entirely
- C. using only the largest outcome

Q72. A 'mutually exclusive and exhaustive' set of scenarios means the scenarios:

- A. cannot occur together and cover all possibilities
- B. can occur together and overlap
- C. exclude some possible outcomes

Q73. The expected value of a constant a times a random variable X , $E(aX)$, equals:

- A. $a \times E(X)$
- B. $a + E(X)$
- C. $E(X) / a$

Q74. A portfolio analyst assigns a 30% probability to a recession. Conditional on recession, a stock returns -8%; otherwise it returns +12%. The expected return is closest to:

- A. +6.0%
- B. +12.0%
- C. -8.0%

Q75. Bayes' formula is MOST useful in investment settings for:

- A. revising the probability of a state given a new signal or observation
- B. computing a bond's yield-to-maturity
- C. measuring the kurtosis of returns

Q76. If two events are mutually exclusive, then $P(A \text{ and } B)$ equals:

- A. 0
- B. $P(A) \times P(B)$
- C. $P(A) + P(B)$

Q77. The variance of a random variable can be computed using a probability tree by:

- A. probability-weighting the squared deviations of each terminal outcome from $E(X)$
- B. summing all terminal outcomes
- C. averaging the probabilities

Q78. Given $P(B) = 0.5$ and $P(AB) = 0.2$, the conditional probability $P(A|B)$ is:

- A. 0.40
- B. 0.10
- C. 0.70

Q79. In Bayes' formula, the denominator (unconditional probability of the new information) is typically computed using the:

- A. total probability rule
- B. harmonic mean
- C. central limit theorem

Q80. The standard deviation of a random variable is always:

- A. non-negative

- B. negative for risky assets
- C. equal to the variance

Module 5: Portfolio Mathematics

20 questions

- Q81.** The expected return of a portfolio is BEST described as:
- A. a weighted average of the expected returns on the component securities
 - B. the simple sum of the component security returns
 - C. the geometric mean of component returns
- Q82.** A portfolio invests 60% in Asset A (expected return 8%) and 40% in Asset B (expected return 12%). The portfolio expected return is:
- A. 9.6%
 - B. 10.0%
 - C. 20.0%
- Q83.** Portfolio variance depends on the risk of individual assets and their:
- A. covariances
 - B. expected returns only
 - C. weights only
- Q84.** The covariance between the returns of two assets can be:
- A. only positive
 - B. negative, zero, or positive
 - C. only between -1 and +1
- Q85.** For a two-asset portfolio, the variance is given by $w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2$ plus:
- A. $2 w_1 w_2 \text{Cov}(R_1, R_2)$
 - B. $w_1 w_2 (\sigma_1 + \sigma_2)$
 - C. $\text{Cov}(R_1, R_2) / (w_1 w_2)$
- Q86.** The correlation coefficient between two assets is computed as the covariance divided by:
- A. the product of the two standard deviations
 - B. the sum of the two variances
 - C. the portfolio mean
- Q87.** Diversification benefits in a two-asset portfolio are GREATEST when the correlation between the assets is:
- A. +1
 - B. 0
 - C. -1
- Q88.** If two assets have a correlation of +1, the portfolio standard deviation is:
- A. the weighted average of the two standard deviations
 - B. always zero
 - C. always greater than either asset's standard deviation
- Q89.** Two random variables X and Y are independent if and only if their joint probability function satisfies:
- A. $P(X, Y) = P(X)P(Y)$
 - B. $P(X, Y) = P(X) + P(Y)$
 - C. $P(X, Y) = P(X)/P(Y)$
- Q90.** For two independent (and therefore uncorrelated) random variables, the expected value of their product equals:

- A. the product of their expected values
- B. the sum of their expected values
- C. zero

Q91. Shortfall risk is BEST defined as the:

- A. probability that a portfolio's value will fall below a minimum acceptable level
- B. maximum possible loss on a portfolio
- C. variance of portfolio returns

Q92. Roy's safety-first criterion selects the portfolio that:

- A. maximizes the safety-first ratio (SFRatio)
- B. minimizes the expected return
- C. maximizes the variance

Q93. The safety-first ratio (SFRatio) is computed as:

- A. $[E(R_p) - R_L] / \sigma_p$
- B. $[E(R_p) + R_L] / \sigma_p$
- C. $\sigma_p / [E(R_p) - R_L]$

Q94. Portfolio P has $E(R) = 10\%$, threshold return 2% , and $SD = 16\%$. Its safety-first ratio is:

- A. 0.50
- B. 0.75
- C. 0.625

Q95. Assuming returns are normally distributed, the optimal safety-first portfolio also minimizes the:

- A. probability of returns below the threshold level
- B. portfolio expected return
- C. number of assets held

Q96. Covariance of an asset's return with itself equals the asset's:

- A. variance
- B. standard deviation
- C. expected return

Q97. Adding an asset with low or negative correlation to a portfolio will MOST likely:

- A. reduce the portfolio's overall risk
- B. increase the portfolio's overall risk
- C. leave portfolio risk unchanged

Q98. Using a joint probability function, the covariance of returns R_A and R_B is computed by summing, over all outcomes:

- A. $P(R_{A,i}, R_{B,j}) \times (R_{A,i} - E[R_A]) \times (R_{B,j} - E[R_B])$
- B. $P(R_{A,i}) \times P(R_{B,j})$
- C. $(R_{A,i} - R_{B,j})^2$

Q99. If the safety-first ratios of three portfolios are 0.40, 0.55, and 0.50, an investor following Roy's criterion should choose the portfolio with SFRatio:

- A. 0.55
- B. 0.50
- C. 0.40

Q100. A portfolio's expected return calculation requires which inputs?

- A. The expected returns of the assets and their portfolio weights

- B. Only the covariances of the assets
- C. Only the variances of the assets

Module 6: Simulation Methods

20 questions

- Q101.** A random variable Y follows a lognormal distribution if:
- A. the natural logarithm of Y is normally distributed
 - B. Y is normally distributed
 - C. Y is uniformly distributed
- Q102.** The lognormal distribution is widely used to model asset PRICES primarily because it is:
- A. bounded below by zero and skewed to the right
 - B. symmetric and can be negative
 - C. always thin-tailed
- Q103.** The lognormal distribution is typically:
- A. skewed to the right (long right tail)
 - B. skewed to the left
 - C. perfectly symmetric
- Q104.** Which distribution is often a good approximation for asset RETURNS (as opposed to prices)?
- A. The normal distribution
 - B. The lognormal distribution
 - C. The chi-square distribution
- Q105.** Continuously compounded returns play a key role in many asset pricing models and are MOST naturally linked to which distribution of prices?
- A. Lognormal
 - B. Uniform
 - C. Binomial
- Q106.** Monte Carlo simulation is MOST commonly used to:
- A. value securities with complex features where no analytic pricing formula exists
 - B. compute a simple arithmetic mean
 - C. rank categorical data
- Q107.** A Monte Carlo simulation works by:
- A. generating a large number of random samples from specified probability distributions
 - B. using only historical average returns
 - C. solving a single deterministic equation
- Q108.** A key limitation of Monte Carlo simulation is that:
- A. its results are only as good as the assumptions and distributions specified
 - B. it can only be used for bonds
 - C. it requires no input assumptions
- Q109.** Bootstrap resampling constructs a sampling distribution by:
- A. repeatedly drawing samples from the observed sample, treating it as the population
 - B. drawing samples from a theoretical normal distribution only
 - C. using a single draw from the population
- Q110.** Bootstrapping is justified by the idea that:
- A. a random sample offers a good representation of the population

- B. the population is always normally distributed
- C. sample size does not matter

Q111. A primary difference between Monte Carlo simulation and bootstrapping is that bootstrapping:

- A. resamples from observed historical data rather than a specified theoretical distribution
- B. never uses random sampling
- C. cannot be used for standard errors

Q112. Monte Carlo simulation is BEST described as a complement to, rather than a replacement for, which type of analysis?

- A. Analytical (closed-form) methods
- B. Categorical data analysis
- C. Ranking methods

Q113. In a Monte Carlo simulation of an option, repeatedly simulating the underlying asset's price path allows estimation of the option's:

- A. expected payoff and hence its value
- B. exact historical return
- C. credit rating

Q114. Bootstrapping can be applied in finance for all of the following EXCEPT:

- A. estimating the standard error of a sample mean
- B. historical simulation in asset allocation
- C. guaranteeing future investment returns

Q115. If continuously compounded returns are normal with mean μ and variance σ^2 , the corresponding price distribution is lognormal because prices equal:

- A. the initial price times e raised to the return
- B. the initial price plus the return
- C. the return divided by the initial price

Q116. A Monte Carlo simulation that produces a wide range of possible outcomes is MOST useful for:

- A. assessing the risk and distribution of potential results
- B. finding a single guaranteed outcome
- C. eliminating all uncertainty

Q117. The lognormal distribution appears in which well-known option pricing model?

- A. Black-Scholes-Merton
- B. Gordon growth model
- C. Capital asset pricing model

Q118. Compared with the normal distribution, the lognormal distribution CANNOT take:

- A. negative values
- B. values above one
- C. fractional values

Q119. Which step is essential when designing a Monte Carlo simulation?

- A. Specifying the probability distribution(s) for the risk factors
- B. Assuming all variables are constant
- C. Using only one random draw

Q120. Bootstrap resampling typically draws samples that are:

- A. the same size as the original observed sample

- B. always larger than the population
- C. always a single observation

Module 7: Estimation and Inference

20 questions

Q121. Which of the following is a PROBABILITY sampling method?

- A. Simple random sampling
- B. Convenience sampling
- C. Judgmental sampling

Q122. Stratified random sampling involves:

- A. dividing the population into subgroups and randomly sampling from each
- B. selecting whichever observations are easiest to obtain
- C. relying on the researcher's expert judgment

Q123. Sampling error is BEST defined as the difference between:

- A. a sample statistic and the population parameter it estimates
- B. two different population parameters
- C. the mean and the median of a sample

Q124. A key advantage of non-probability sampling methods is that they:

- A. allow quick and low-cost data collection
- B. guarantee a representative sample
- C. eliminate sampling error

Q125. A major risk of non-probability sampling is that the sample may be:

- A. non-representative of the population
- B. too large to analyze
- C. free of any bias

Q126. The central limit theorem states that, for large sample size n , the sampling distribution of the sample mean is approximately:

- A. normal with mean μ and variance σ^2/n
- B. lognormal with mean μ
- C. uniform regardless of n

Q127. A powerful implication of the central limit theorem is that it holds:

- A. regardless of the shape of the underlying population distribution
- B. only when the population is normal
- C. only for binomial populations

Q128. The standard error of the sample mean equals the population standard deviation divided by:

- A. the square root of n
- B. n
- C. n squared

Q129. If a population has a standard deviation of 20 and the sample size is 100, the standard error of the sample mean is:

- A. 2.0
- B. 0.2
- C. 20.0

Q130. All else equal, increasing the sample size n will cause the standard error of the sample mean to:

- A. decrease
- B. increase
- C. remain unchanged

Q131. According to the central limit theorem, the mean of the sampling distribution of the sample mean equals the:

- A. population mean
- B. sample standard deviation
- C. median of the sample

Q132. Resampling methods such as the bootstrap and jackknife are used to:

- A. estimate the sampling distribution of a statistic
- B. eliminate the need for any data
- C. compute the population parameter exactly

Q133. The bootstrap differs from a purely analytical approach to the standard error in that it:

- A. uses computer simulation rather than an analytical formula such as a z- or t-statistic
- B. requires the population variance to be known
- C. can only be used for the mean

Q134. A potential advantage of the bootstrap over analytical formulas is its:

- A. accuracy and wide applicability to complicated estimators
- B. requirement of fewer than ten observations
- C. guarantee of zero sampling error

Q135. Which sampling method selects observations based purely on ease of access?

- A. Convenience sampling
- B. Stratified random sampling
- C. Simple random sampling

Q136. In simple random sampling, each member of the population has:

- A. an equal chance of being selected
- B. a chance proportional to its value
- C. no chance of being selected twice

Q137. The standard error is important because it measures the:

- A. precision (variability) of the sample mean as an estimate of the population mean
- B. skewness of the population
- C. kurtosis of the sample

Q138. As the number of compounding subsamples grows, bootstrapping is MOST appropriately described as:

- A. repeatedly drawing samples of the same size from the original sample
- B. drawing one large sample from the population
- C. computing a deterministic formula

Q139. Judgmental sampling relies primarily on:

- A. the researcher's expertise to select the sample
- B. a random number generator
- C. stratifying the population

Q140. A larger sample size is generally desirable in estimation because it tends to:

- A. reduce sampling error and increase estimate precision
- B. increase the standard error

C. guarantee the sample is biased

Module 8: Hypothesis Testing

20 questions

- Q141.** Which of the following is the FIRST step in hypothesis testing?
- A. State the hypotheses
 - B. Calculate the test statistic
 - C. Make a decision
- Q142.** A Type I error occurs when an analyst:
- A. rejects a true null hypothesis
 - B. fails to reject a false null hypothesis
 - C. accepts the alternative when it is true
- Q143.** A Type II error occurs when an analyst:
- A. fails to reject a false null hypothesis
 - B. rejects a true null hypothesis
 - C. correctly rejects a false null
- Q144.** The level of significance of a test is the probability of:
- A. a Type I error
 - B. a Type II error
 - C. a correct decision
- Q145.** The power of a test is BEST defined as:
- A. the probability of correctly rejecting a false null hypothesis
 - B. the probability of a Type I error
 - C. the probability of a Type II error
- Q146.** The confidence level of a test is the complement of the:
- A. level of significance
 - B. power of the test
 - C. probability of a Type II error
- Q147.** For a test of the population mean of a normally distributed population with UNKNOWN variance, the theoretically correct test statistic is the:
- A. t-statistic
 - B. z-statistic
 - C. chi-square statistic
- Q148.** A test statistic is BEST described as a quantity, calculated from a sample, that is:
- A. the basis for deciding whether to reject the null hypothesis
 - B. always equal to the population parameter
 - C. the probability of the null being true
- Q149.** To test whether the difference between two means from independent samples (normal populations, unknown but equal variances) is significant, the appropriate test is a:
- A. t-test based on pooling the two samples' observations
 - B. chi-square test
 - C. paired comparisons test
- Q150.** When two samples are dependent (e.g., before-and-after observations on the same subjects), the

appropriate test is a:

- A. paired comparisons (mean differences) test
- B. pooled-variance t-test for independent samples
- C. F-test of variances

Q151. For a test concerning the variance of a SINGLE normally distributed population, the appropriate test statistic is:

- A. chi-square with $n - 1$ degrees of freedom
- B. an F-statistic
- C. a z-statistic

Q152. For a test of the EQUALITY of variances of two normally distributed populations from independent samples, the appropriate test statistic is:

- A. an F-statistic (ratio of sample variances)
- B. a chi-square statistic
- C. a t-statistic

Q153. The degrees of freedom for the F-test comparing two sample variances are:

- A. $n_1 - 1$ and $n_2 - 1$
- B. $n_1 + n_2 - 2$
- C. $n - 1$

Q154. A parametric test is BEST described as a test that:

- A. concerns a population parameter or relies on specific distributional assumptions
- B. makes no distributional assumptions
- C. can only be used on ranked data

Q155. If the calculated test statistic exceeds the critical value (in absolute terms for a two-sided test), the analyst should:

- A. reject the null hypothesis
- B. fail to reject the null hypothesis
- C. increase the significance level

Q156. Decreasing the significance level (α) from 0.05 to 0.01, holding all else constant, will:

- A. decrease the probability of a Type I error but increase the probability of a Type II error
- B. decrease both Type I and Type II error probabilities
- C. increase the probability of a Type I error

Q157. A p-value is BEST described as the:

- A. smallest significance level at which the null hypothesis can be rejected
- B. probability that the null hypothesis is true
- C. probability of a Type II error

Q158. A one-sided (one-tailed) alternative hypothesis is appropriate when the analyst:

- A. is testing for a difference in a specific direction
- B. is testing only for any difference, regardless of direction
- C. has no prior expectation

Q159. In hypothesis testing, the null hypothesis typically represents the:

- A. statement to be tested, often 'no effect' or 'no difference'
- B. outcome the researcher hopes to prove
- C. alternative under consideration

Q160. For a fixed sample size, increasing the power of a test generally requires:

- A. accepting a higher probability of a Type I error
- B. reducing the significance level
- C. no trade-off whatsoever

Module 9: Parametric and Non-Parametric Tests of Independence

20 questions

- Q161.** A two-sided alternative hypothesis concerning a parameter θ is BEST written as:
- A. $H_0: \theta = \theta_0$ versus $H_a: \theta \neq \theta_0$
 - B. $H_0: \theta \leq \theta_0$ versus $H_a: \theta > \theta_0$
 - C. $H_0: \theta \geq \theta_0$ versus $H_a: \theta < \theta_0$
- Q162.** A nonparametric test is MOST appropriate when:
- A. data do not meet distributional assumptions or are given in ranks
 - B. the population is known to be normal
 - C. a population parameter is the sole focus and assumptions hold
- Q163.** To test whether a population correlation coefficient differs from zero, the test statistic follows a t-distribution with degrees of freedom equal to:
- A. $n - 2$
 - B. $n - 1$
 - C. n
- Q164.** The null hypothesis for the most common two-sided test of correlation is:
- A. $H_0: \rho = 0$
 - B. $H_0: \rho = 1$
 - C. $H_0: \rho < 0$
- Q165.** The Spearman rank correlation coefficient is calculated using the:
- A. ranks of the two variables within their samples
 - B. raw values of the two variables
 - C. squared values of the variables
- Q166.** The Spearman rank correlation is MOST useful when:
- A. data do not meet the assumptions required for the parametric correlation test
 - B. the data are perfectly normally distributed
 - C. the sample size exceeds one million
- Q167.** A test of independence between two categorical variables uses which test statistic?
- A. Chi-square
 - B. t-statistic
 - C. F-statistic
- Q168.** A test of independence using a contingency table compares actual frequencies with the frequencies expected under:
- A. the assumption of independence
 - B. perfect positive correlation
 - C. a normal distribution
- Q169.** A contingency table is BEST described as a table that:
- A. displays the joint frequencies of two categorical variables
 - B. lists the present values of cash flows
 - C. ranks observations from smallest to largest
- Q170.** For a chi-square test of independence with r rows and c columns, the expected frequency in a cell equals:

- A. (row total x column total) / grand total
- B. row total x column total
- C. (row total + column total) / 2

Q171. A nonparametric test is characterized by the fact that it:

- A. makes minimal assumptions about the population distribution
- B. always requires a normal population
- C. can only test the population mean

Q172. A parametric test, in contrast to a nonparametric test, is one that:

- A. concerns a parameter or relies on specific distributional assumptions
- B. ignores the population distribution
- C. can only use ranked data

Q173. If the calculated chi-square statistic in a test of independence exceeds the critical value, the analyst concludes that the two variables are:

- A. not independent (i.e., related)
- B. independent
- C. perfectly correlated

Q174. Which situation MOST strongly favors a nonparametric over a parametric test?

- A. The presence of significant outliers in small samples
- B. A large sample drawn from a normal population
- C. A test concerning a single normal mean with known variance

Q175. A right-side one-sided test of correlation has the alternative hypothesis:

- A. $H_a: \rho > 0$
- B. $H_a: \rho < 0$
- C. $H_a: \rho \text{ not equal to } 0$

Q176. The t-statistic for testing a correlation coefficient increases (in absolute value), all else equal, as the sample size n:

- A. increases
- B. decreases
- C. stays the same

Q177. A contingency-table chi-square test statistic is computed by summing, across all cells, the quantity:

- A. $(\text{observed} - \text{expected})^2 / \text{expected}$
- B. $(\text{observed} - \text{expected}) / \text{observed}$
- C. observed x expected

Q178. Testing whether two categorical variables (e.g., sector and credit-rating category) are related is BEST handled with a:

- A. chi-square test of independence
- B. paired comparisons t-test
- C. Gordon growth model

Q179. If a correlation test fails to reject $H_0: \rho = 0$, the analyst concludes that:

- A. the observed correlation could plausibly be due to chance
- B. the variables are definitely strongly correlated
- C. the correlation equals exactly +1

Q180. Both parametric and nonparametric tests of correlation aim to assess whether a relationship between two

variables:

- A. is the result of chance
- B. is caused by inflation
- C. violates the no-arbitrage condition

Module 10: Simple Linear Regression

20 questions

- Q181.** In a simple linear regression, the variable we wish to explain is the:
- A. dependent variable
 - B. independent variable
 - C. residual
- Q182.** Simple linear regression estimates the slope and intercept using which approach?
- A. Ordinary least squares
 - B. Maximum range
 - C. Geometric averaging
- Q183.** The least squares criterion minimizes the sum of the:
- A. squared vertical distances between observations and the regression line
 - B. absolute slopes of the line
 - C. values of the independent variable
- Q184.** Which of the following is NOT one of the four assumptions of the simple linear regression model?
- A. Linearity
 - B. Homoskedasticity
 - C. Multicollinearity
- Q185.** Homoskedasticity refers to the assumption that the:
- A. variance of the residuals is constant across observations
 - B. residuals are perfectly correlated
 - C. independent variable is random
- Q186.** The independence assumption in simple linear regression means that the:
- A. X-Y pairs are uncorrelated across observations
 - B. intercept equals zero
 - C. residuals have non-constant variance
- Q187.** The normality assumption in simple linear regression requires that the:
- A. residuals be normally distributed
 - B. raw data themselves be normally distributed
 - C. independent variable be normally distributed
- Q188.** The total variation in the dependent variable (SST) can be decomposed into:
- A. explained variation (SSR) plus unexplained variation (SSE)
 - B. the slope plus the intercept
 - C. the mean plus the median
- Q189.** The coefficient of determination (R-squared) is BEST interpreted as the:
- A. proportion of the total variation in Y explained by the regression
 - B. slope of the regression line
 - C. standard error of the estimate
- Q190.** In a simple linear regression, the coefficient of determination equals the:
- A. square of the correlation coefficient between X and Y
 - B. correlation coefficient itself

C. slope coefficient

Q191. The standard error of the estimate (SEE) measures the:

- A. standard deviation of the regression residuals
- B. slope of the regression
- C. total variation in X

Q192. A smaller standard error of the estimate indicates that the regression:

- A. fits the data more closely
- B. fits the data less closely
- C. has a larger intercept

Q193. Analysis of variance (ANOVA) in regression is used to:

- A. evaluate the overall fit and significance of the regression
- B. compute the present value of cash flows
- C. rank the residuals

Q194. The F-statistic in a simple linear regression tests whether the:

- A. regression as a whole is statistically significant (slope differs from zero)
- B. residuals are normally distributed
- C. intercept equals the mean of Y

Q195. To test whether a regression slope coefficient is statistically different from zero, the analyst uses a:

- A. t-test on the estimated slope
- B. chi-square test
- C. test of the coefficient of variation

Q196. In the regression $Y = b_0 + b_1X$, the slope coefficient b_1 represents the:

- A. change in Y for a one-unit change in X
- B. value of Y when X is zero
- C. correlation between X and Y

Q197. In the regression $Y = b_0 + b_1X$, the intercept b_0 represents the:

- A. expected value of Y when X equals zero
- B. change in Y per unit change in X
- C. standard error of the estimate

Q198. A prediction interval for a forecasted value of the dependent variable accounts for:

- A. uncertainty in the regression model and the residual variability
- B. only the slope coefficient
- C. the mean of X alone

Q199. Residual plots are used in regression analysis primarily to:

- A. detect violations of the regression assumptions
- B. compute the dividend growth rate
- C. rank the independent variables

Q200. A pattern (e.g., seasonality) in a plot of regression residuals over time MOST likely indicates a violation of the assumption of:

- A. independence (suggesting autocorrelation)
- B. linearity
- C. a zero intercept

Module 11: Introduction to Big Data Techniques

20 questions

- Q201.** Big Data is commonly characterized by the 'three Vs', which are:
- A. volume, velocity, and variety
 - B. value, volatility, and volume
 - C. velocity, validity, and value
- Q202.** When Big Data is used for inference or prediction, an important fourth 'V' to consider is:
- A. veracity
 - B. volatility
 - C. value
- Q203.** The term 'fintech' MOST broadly refers to:
- A. technology-driven innovation in the financial services industry
 - B. a specific type of government bond
 - C. a method of computing portfolio variance
- Q204.** Which of the following are among the main sources of ALTERNATIVE data?
- A. Individuals, business processes, and sensors
 - B. Only corporate financial statements
 - C. Only government economic releases
- Q205.** Artificial intelligence (AI) is BEST described as computer systems capable of:
- A. performing tasks that traditionally required human intelligence
 - B. only storing large datasets
 - C. executing only fixed, pre-programmed rules
- Q206.** Machine learning (ML) is BEST summarized by the phrase:
- A. 'find the pattern, apply the pattern'
 - B. 'buy low, sell high'
 - C. 'minimize the sum of squared errors'
- Q207.** Which of the following is a main TYPE of machine learning?
- A. Supervised learning
 - B. Convenience learning
 - C. Judgmental learning
- Q208.** In supervised learning, the algorithm is trained using:
- A. labeled data (inputs paired with known outputs)
 - B. only unlabeled data
 - C. no data at all
- Q209.** Unsupervised learning differs from supervised learning in that it:
- A. works with unlabeled data to find structure or groupings
 - B. always requires labeled outputs
 - C. can only be used for bond pricing
- Q210.** Natural language processing (NLP) is an application of text analytics that:
- A. analyzes and interprets text- and voice-based data using the structure of human language
 - B. computes the variance of returns

C. ranks bonds by credit quality only

Q211. Deep learning is BEST described as a type of:

- A. machine learning
- B. fixed rule-based system
- C. sampling method

Q212. Among the developments most relevant to quantitative investment analysis is the:

- A. analysis of large traditional and alternative datasets
- B. elimination of all financial data
- C. exclusive use of paper records

Q213. Alternative data generated by SENSORS would MOST likely include:

- A. satellite imagery or Internet-of-Things device readings
- B. a company's audited income statement
- C. a central bank interest-rate announcement

Q214. Alternative data generated by INDIVIDUALS would MOST likely include:

- A. social media posts and online reviews
- B. corporate 10-K filings
- C. exchange-reported trading volumes

Q215. A primary motivation for integrating alternative data into investment processes is to:

- A. help generate alpha and reduce losses
- B. eliminate the need for any analysis
- C. guarantee positive returns

Q216. Early forms of fintech primarily involved:

- A. data processing and automation of routine tasks
- B. complex deep-learning trading systems
- C. natural language processing of voice data

Q217. 'Veracity' as a characteristic of Big Data refers to the data's:

- A. truthfulness, quality, and reliability
- B. speed of arrival
- C. physical storage size

Q218. 'Velocity' as one of the three Vs of Big Data refers to the:

- A. speed at which data are created and become available
- B. total quantity of data
- C. diversity of data formats

Q219. 'Variety' as one of the three Vs of Big Data refers to the:

- A. range of data types and sources, including structured and unstructured data
- B. reliability of the data
- C. rate of data generation

Q220. Fundamental asset managers are described as using Big Data and ML tools to engage in:

- A. hybrid forms of investment decision making
- B. purely manual stock picking only
- C. eliminating all quantitative inputs